Facultad de Ciencias
UNAM

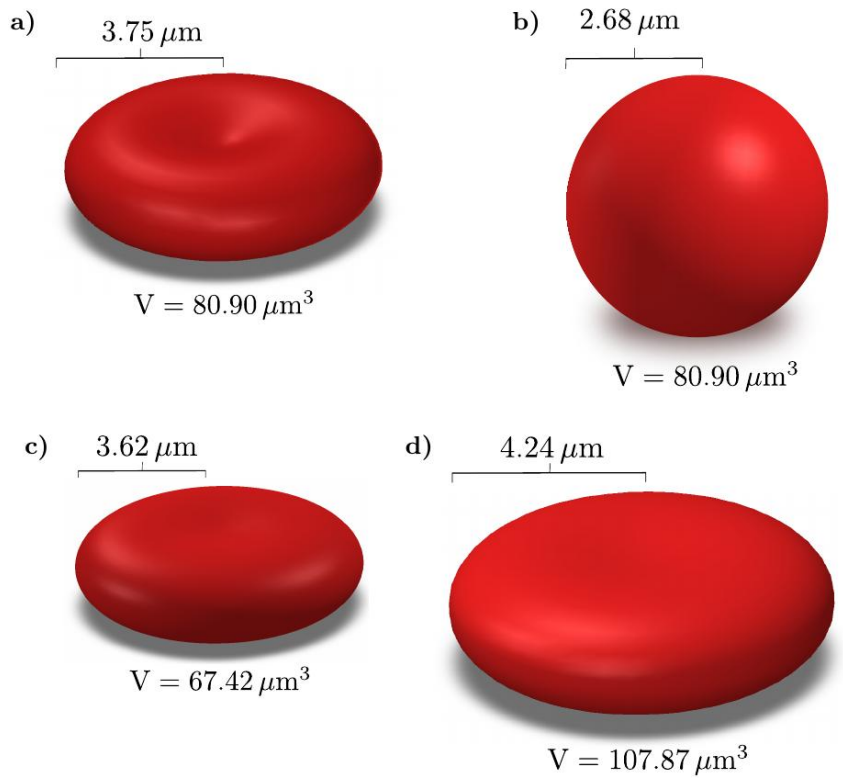
AVANCES DE TESIS DE LICENCIATURA

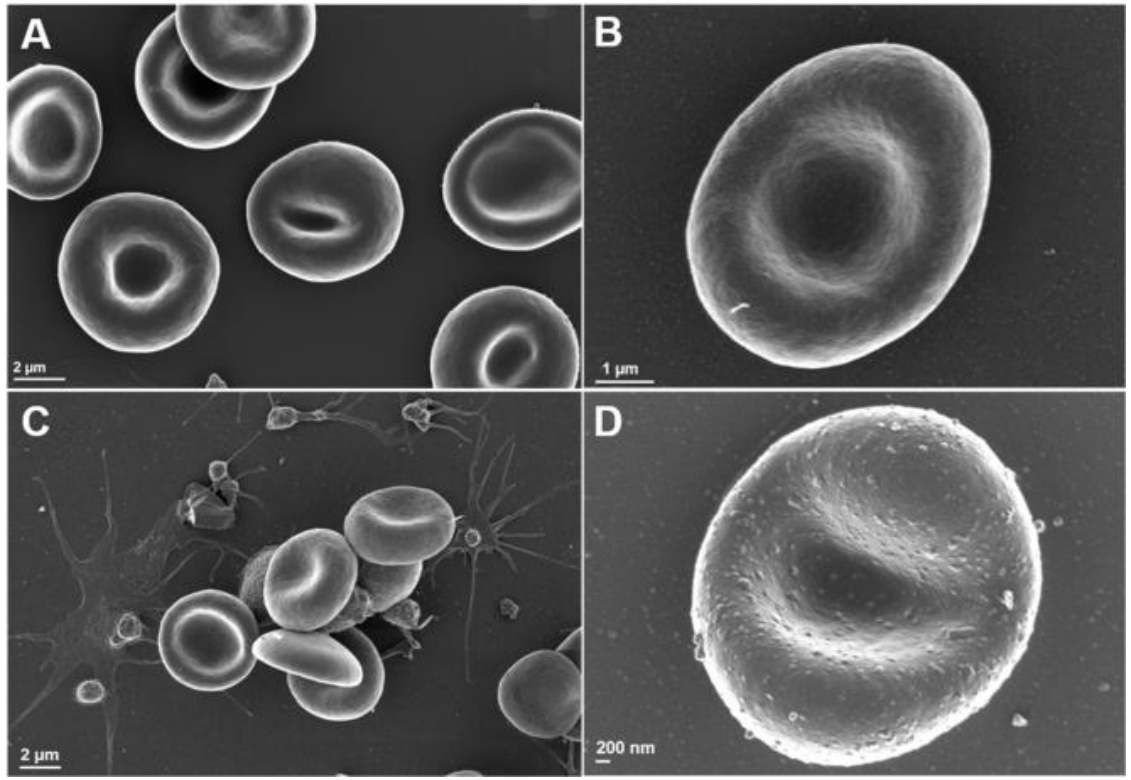
Cálculo numérico de propiedades ópticas de eritrocitos sanos y enfermos

Dana Larissa Luna González

M. en C. Jonathan Alexis Urrutia Anguiano
Dr. Alejandro Reyes Coronado

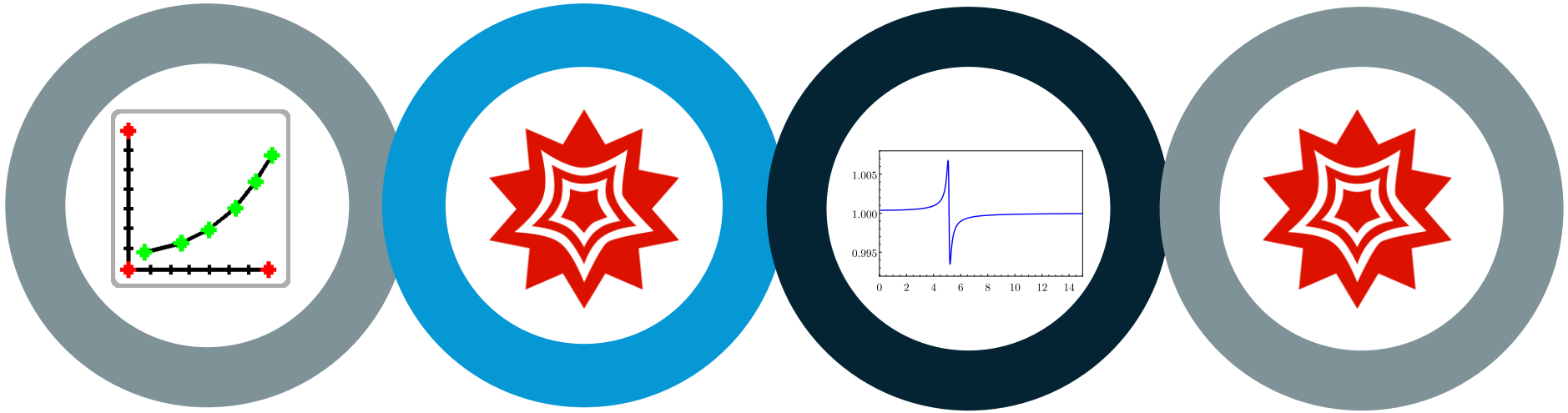
Antecedentes





Alummoottil, M. , *et al.* *Hemato*, **15**(4), 2010

Funciones dieléctricas



Obtención de los datos experimentales

Implementación de las relaciones de Kramers-Kronig en Mathematica

Validación de las relaciones de KK con lorentzianas

Construcción de la parte real y la imaginaria de la función dieléctrica de los eritrocitos

Datos experimentales

$$n(\lambda) = \eta(\lambda) + i\kappa(\lambda)$$

Eritrocitos

Parte real: 250 - 1100 nm

$$\eta_{\mathrm{Hb}}(\lambda, c_{\mathrm{Hb}}) = \eta_{\mathrm{H_2O}}(\lambda)[\beta(\lambda)c_{\mathrm{Hb}} + 1]$$

Friebel, M., y Meinke, M. *Applied Optics*, **45**(12), 2006

Parte imaginaria:

$$\kappa_{\mathrm{Hb}}$$

- 28.7 g/dL 250 - 600 nm
- 30.6 g/dL 250 - 1050 nm
- 34 g/dL 250 - 600 nm

Friebel, M., y Meinke, M. *Applied Optics*, **10**(6), 2005

Plasma

Parte real: 350 - 750 nm

$$\eta_{\mathrm{Plasma}}$$

Liu, S., *et al.* *Journal of Biomedical Optics*, **24**(3), 2019

Parte imaginaria: 350 - 1050 nm

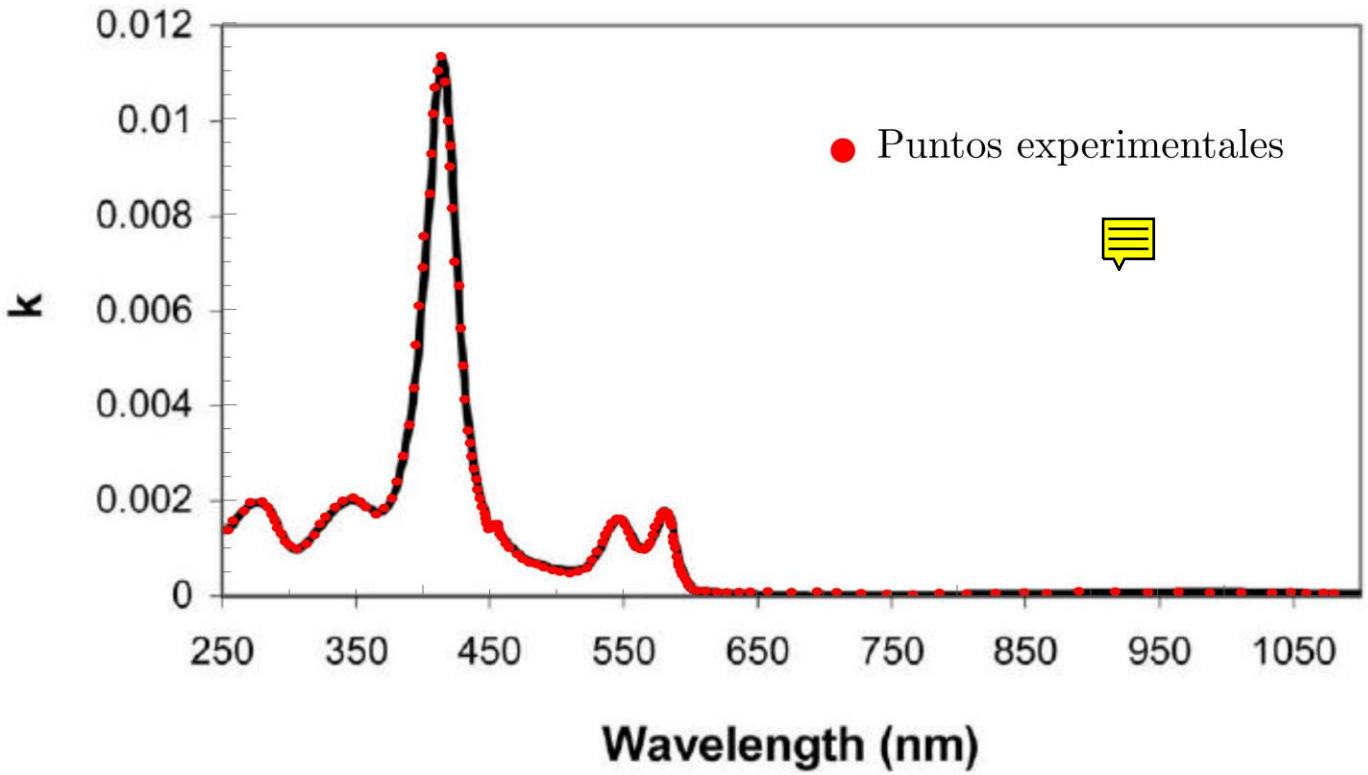
$$\kappa_{\mathrm{Plasma}}$$

Meinke, M., *et al.* *Journal of Biomedical Optics*, **12**(1), 2007

Datos experimentales

Engauge Digitizer

Eritrocitos **28.7 g/dL**



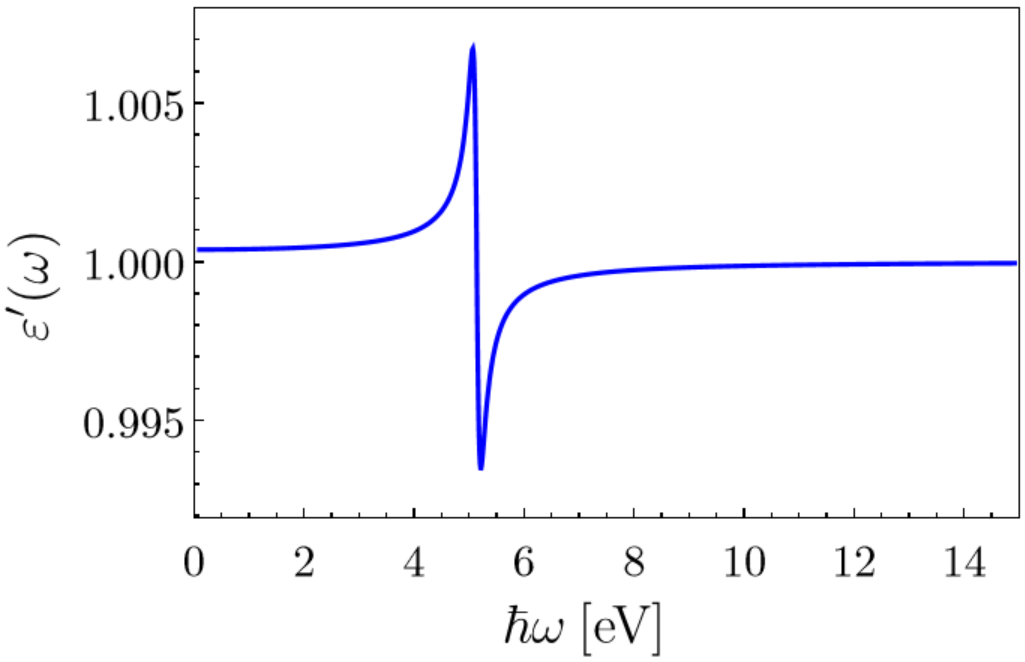
Friebel, M. y Meinke, M. *Applied Optics*, **10**(6), 2005

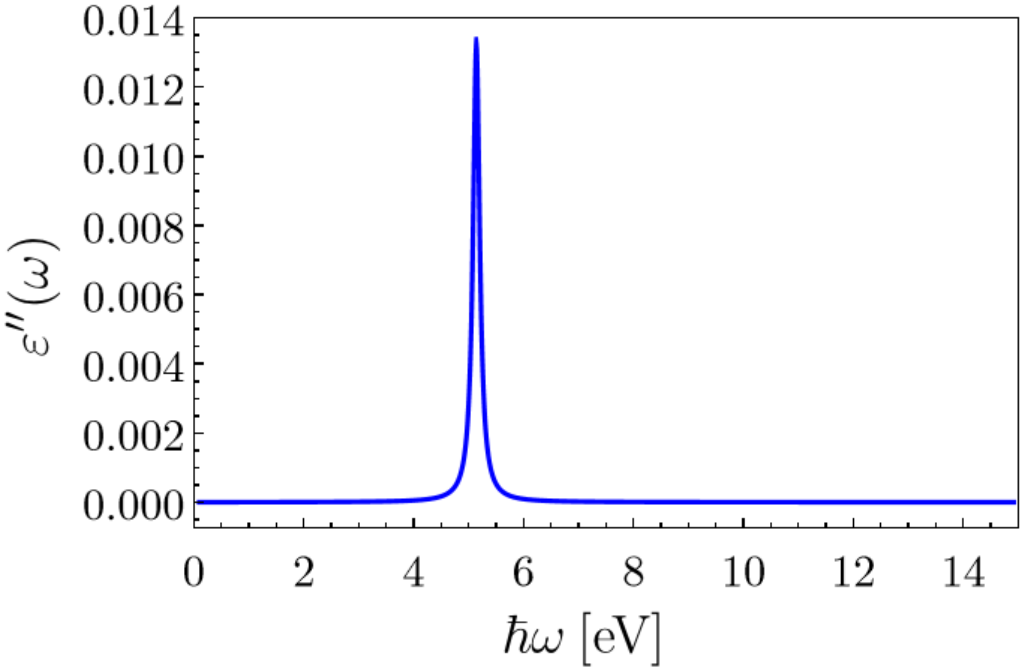
Relaciones de KK

Lorentzianas

$$\varepsilon_{\text{Lorentz}}(\omega) = 1 - \left[\frac{\omega_p^2}{\omega(\omega + I\gamma)}\right]$$

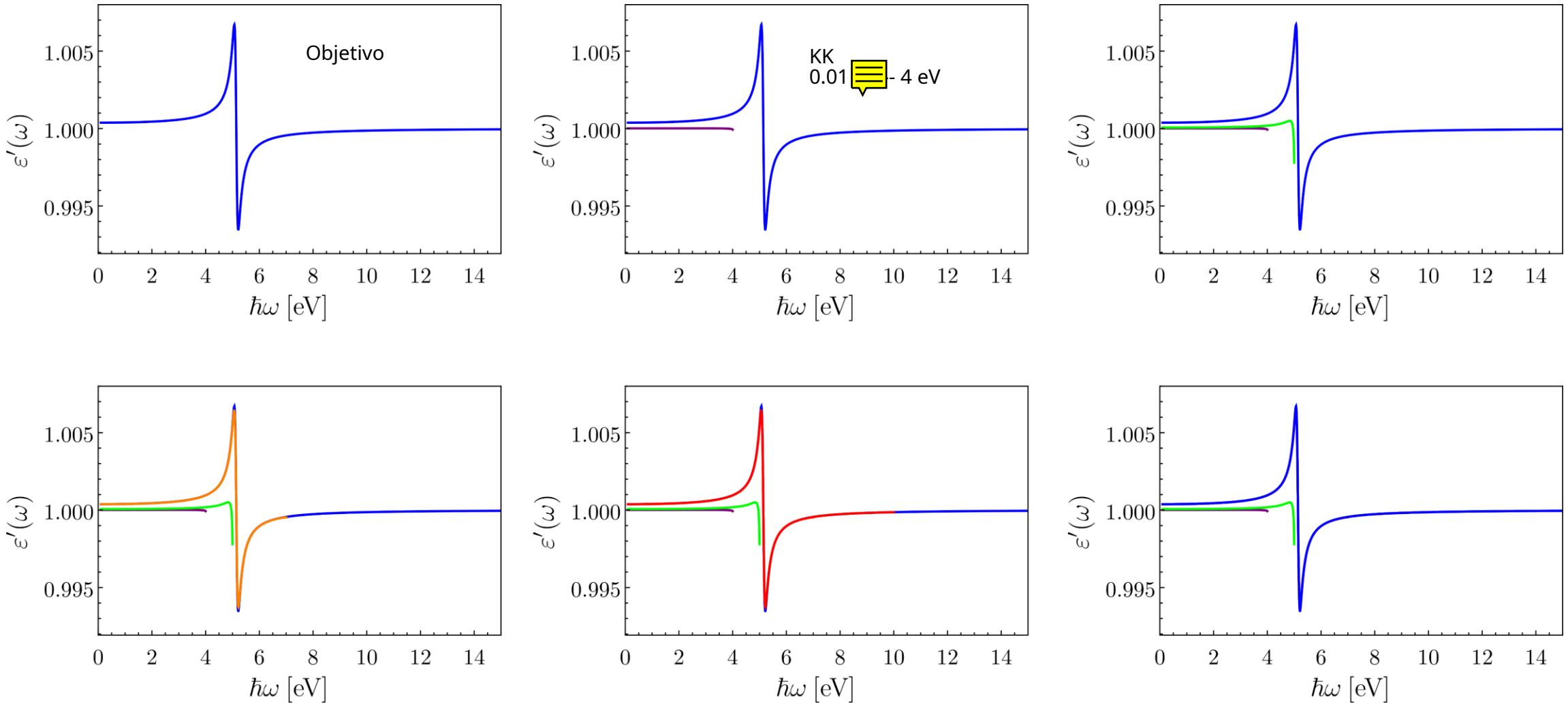
$$A = 2\ \text{eV}^2, \hbar\omega_p = 5.142\ \text{eV}, \hbar\gamma = 0.147\ \text{eV}$$





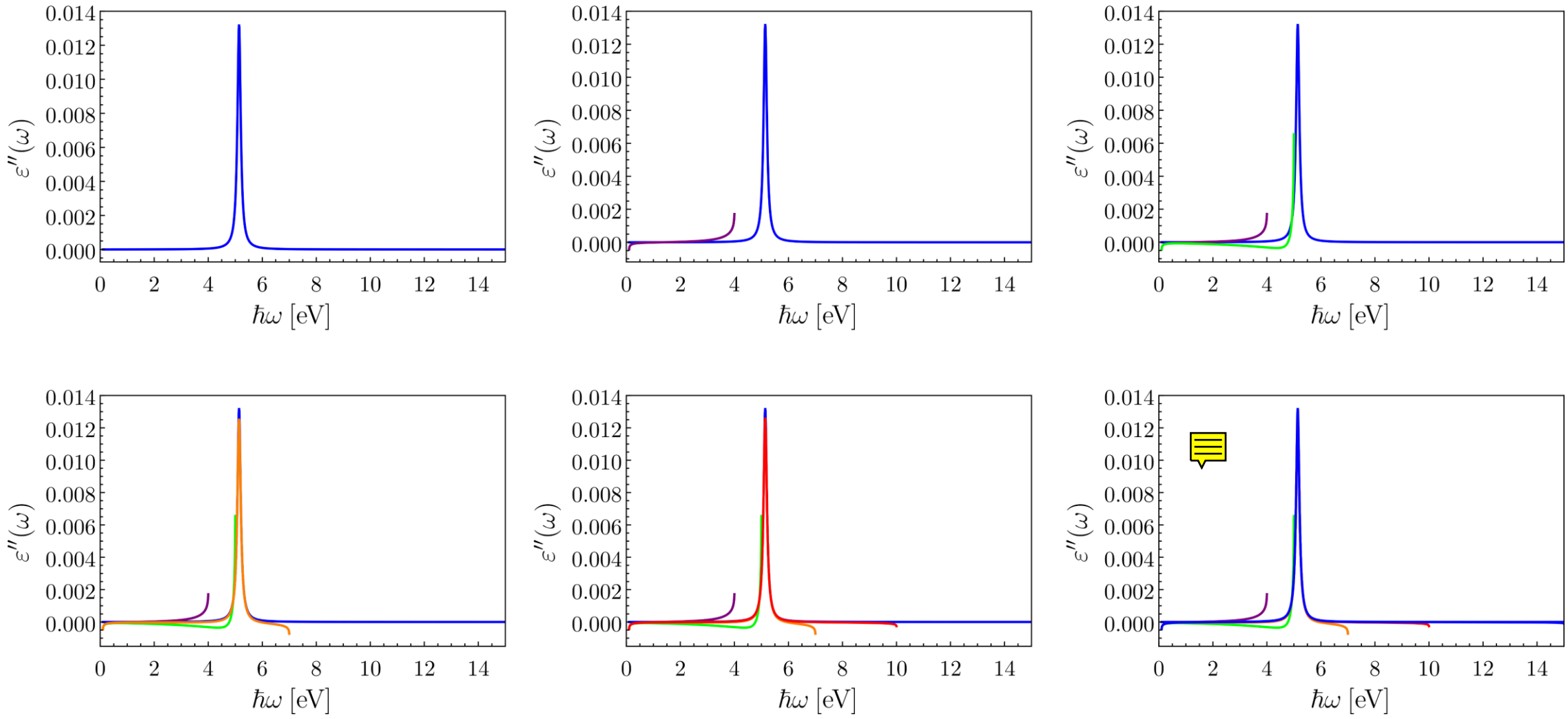
Relaciones de KK

Parte real

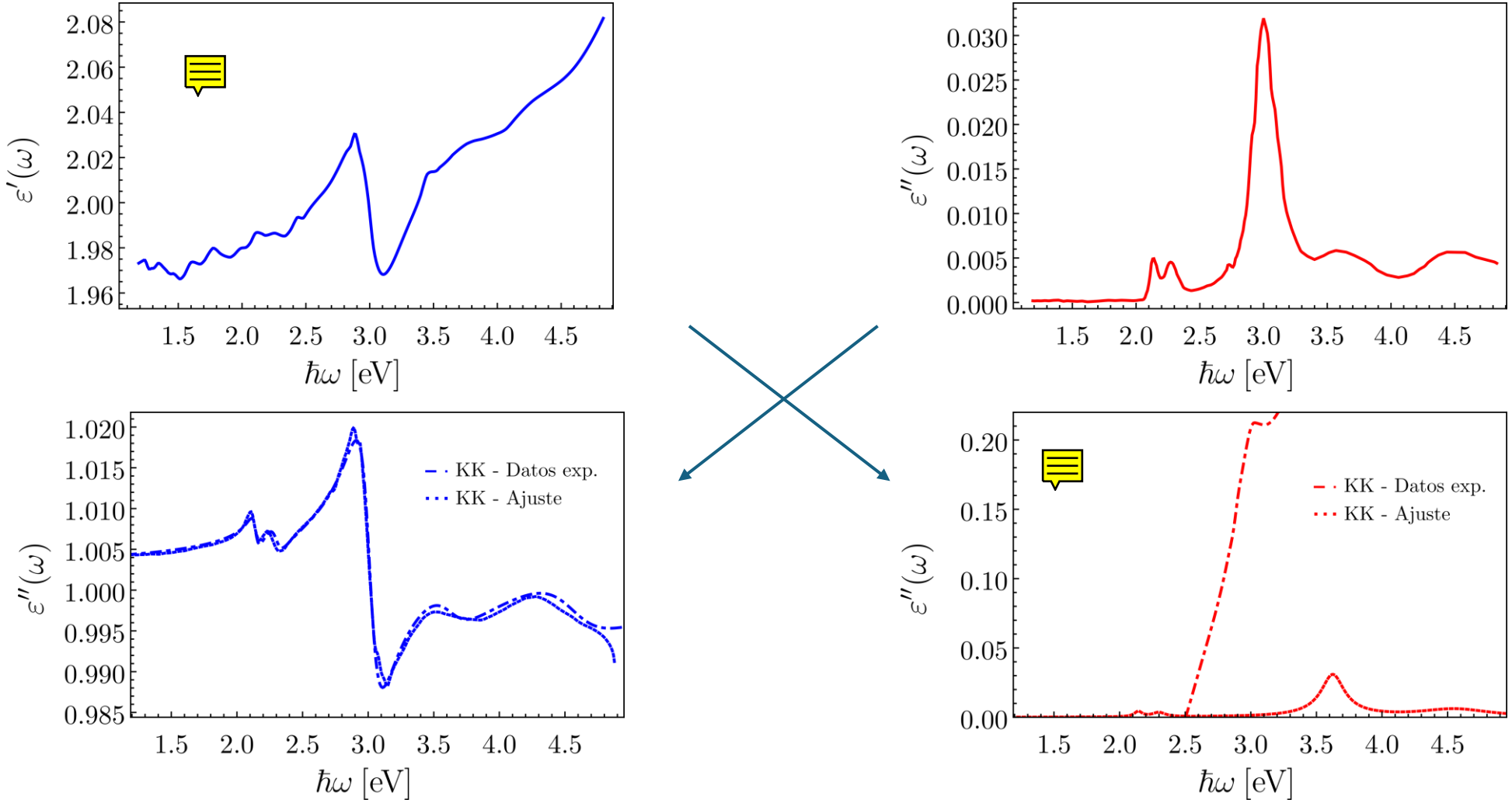


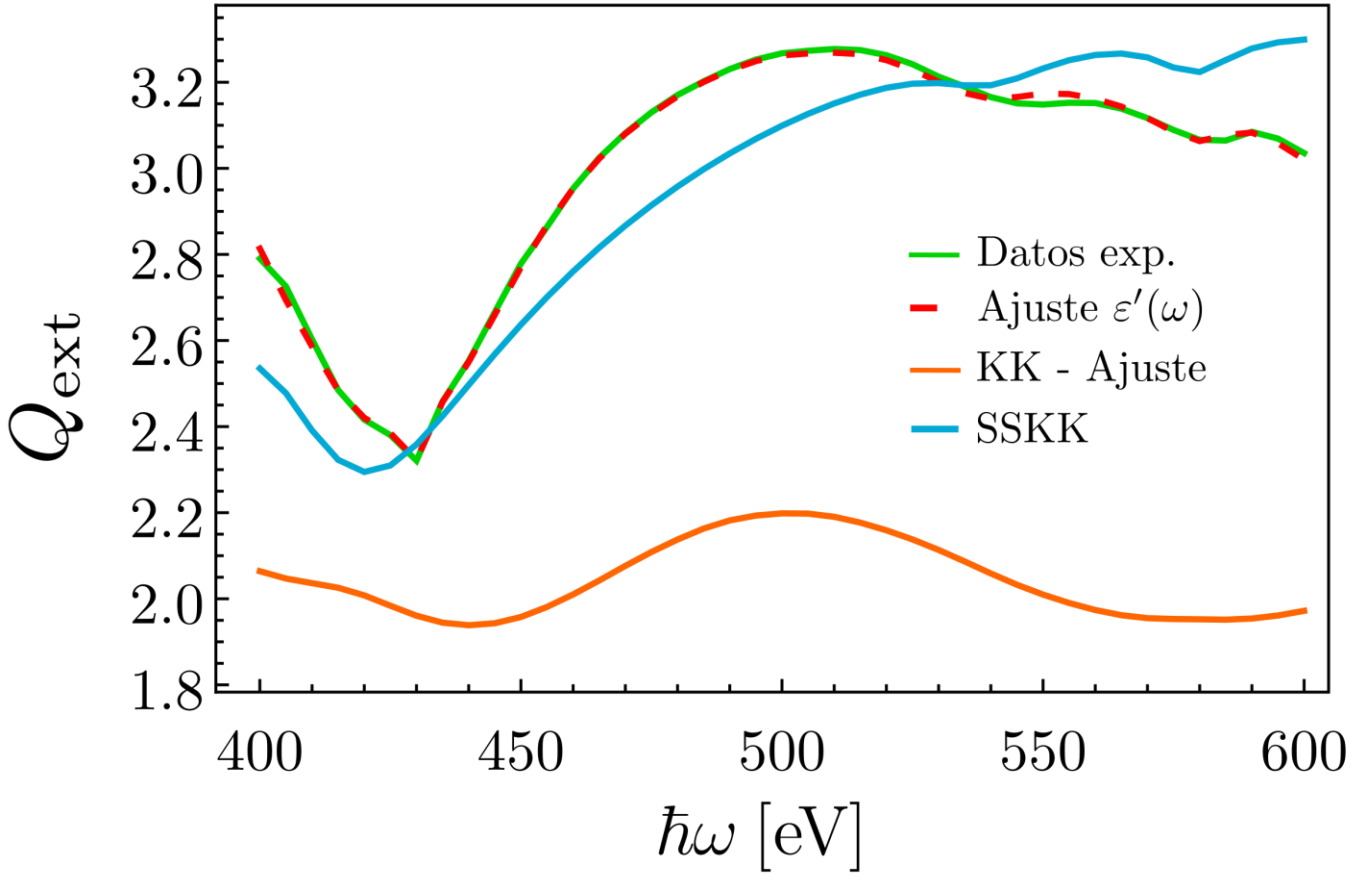
Relaciones de KK

Parte imaginaria



Relaciones de KK



Friebel, M., y Meinke, M. *Applied Optics*, **10**(6), 2005

Secciones transversales